

AMOGH SOMISETTY

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Electrical Engineering student focused on FPGA and digital hardware design for low-latency, high-performance systems. Built an RV32I CPU from the micro-architecture up in SystemVerilog and brought it up on a Xilinx FPGA, closing timing through the full Vivado synthesis, place-and-route, and static-timing flow. Strong at simulation and waveform debug, clean RTL, and high-speed interfacing; drawn to hardware where nanoseconds decide the outcome.

EDUCATION

California Polytechnic State University, San Luis Obispo

B.S. Electrical Engineering

Expected June 2028

San Luis Obispo, CA

- **Coursework:** Computer Architecture, Digital Logic Design, Embedded Systems, Circuits I-III, Signals & Systems

FPGA & DIGITAL DESIGN PROJECTS

RV32I Processor — RTL to FPGA | *SystemVerilog, Xilinx Vivado, Artix-7 (Basys-3)*

- Implemented a multi-cycle RV32I CPU micro-architecture from the RTL up in SystemVerilog: full base ISA, 10-operation ALU, FSM control unit, 32×32 register file, and datapath, coding each block to a defined interface and integrating them into a clean, working core.
- Ran the full FPGA flow in Vivado — synthesis, place & route, and static timing — and iterated on the critical path to meet the target clock on a Xilinx Artix-7, bringing the CPU up and running programs on real hardware.
- Verified the design with self-checking SystemVerilog stimulus and debugged branch-comparison and immediate-generation logic through **waveform analysis**, tracing incorrect results to the exact RTL before integration.

Mixed-Signal DAC Signal Chain on Custom PCB | *AVR, LTspice, Fusion*

- Designed, simulated, and fabricated a binary-weighted 4-bit DAC on a 2-layer PCB driven by register-level AVR firmware; characterized linearity to 62.5 mV/step within ±1 LSB across all 16 codes.

EXPERIENCE

Founder & Lead Engineer

Theta — open-source hardware reliability tooling (independent research)

May 2026 – Present

San Luis Obispo, CA

- Built an automated Python test and CI stack (pytest, lint, static type-checking) in which real hardware-failure incidents are pinned as **regression tests**, and enforced a permutation-null methodology so no result ships without being checked against a null — the verification discipline performance-critical hardware demands.
- Debugged hardware thermal and electrical anomalies methodically from high-rate device telemetry, isolating a genuinely faulty unit that fixed-threshold monitoring missed; the operator independently confirmed the fault by replacing the part months later.
- Shipped and maintain an open-source agent (`pip install runtheta`) that ingests real-time NVML/DCGM telemetry and automates the analysis flow in Python; advised by ex-Intel silicon-validation faculty.

Avionics Electrical Lead — Boards, Interconnect & High-Speed Serial

Poly UAS (Cal Poly Unmanned Aerial Systems)

Apr 2026 – Present

San Luis Obispo, CA

- Own the avionics electrical design end to end: verified the full sensor / power / data interconnect against the flight-controller pinout and captured a **DroneCAN + high-speed-serial** backplane schematic in KiCad (ERC-clean, netlist-verified) to replace a hand-wired harness (CAN termination, per-arm power); also carrier-board and sensor-module design for the team's NVIDIA Jetson AGX Orin edge-AI stack.

Lab Operations Intern — Systems, Automation & Validation

Natera

June 2025 – Sept 2025

Pleasanton, CA

- Wrote Python automation for barcode validation and report retrieval on a high-throughput production line, cutting validation/packaging cycle time from ~20 min to 10–12 min (>40%); ran root-cause failure analysis across electrical, mechanical, and software systems.

TECHNICAL SKILLS

Digital Design & RTL: SystemVerilog/Verilog RTL, RV32I CPU design, FSM control, datapath / ALU / register-file design, clean well-architected modules, simulation & waveform debug, self-checking testbenches

FPGA, EDA & Timing: Xilinx Vivado (synthesis, place & route), FPGA bring-up (Artix-7 / Basys-3), static timing analysis & timing closure, critical-path / latency-driven iteration, LTspice

Interfaces & Systems: high-speed serial and low-level buses (CAN / DroneCAN, SPI, I2C, UART), PCB schematic capture (KiCad, Altium-style flow), oscilloscope / logic-level debug

Automation & Verification: simulation-based RTL verification, regression / characterization suites, CI (pytest, lint, type-check), Tcl (Vivado flow), Bash / shell, Python test automation

Languages & Tools: SystemVerilog/Verilog, Python, C/C++, Tcl, Bash; register-level firmware (AVR, STM32); Git, Linux

LEADERSHIP

President, Poly Engineering Consulting: lead client-facing engineering teams across disciplines

Sept 2024 – Present

Lab Assistant, Cal Poly EPIC: teach embedded systems to precollegiate students

Summer 2026